
TRADE OPENNESS, INCOME INEQUALITY, AND THE WELFARE STATE: NEW EMPIRICAL EVIDENCE

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ABSTRACT

Welfare states do not merely redistribute income after trade shocks arrive—they alter whether those shocks materialize at all. We estimate how government social spending moderates the effect of trade openness on within-country income inequality using a panel of 120 countries (1990–2018) and two-way fixed effects with interaction terms. We find that trade openness has no unconditional average effect on the Gini coefficient ($\beta = 0.007$, $p = 0.589$). However, a one-standard-deviation increase in trade openness raises inequality by 2.1 Gini points in developed countries with weak welfare states, while the same shock leaves inequality unchanged in developed countries with strong welfare states—a full attenuation. In developing countries, trade modestly compresses inequality, and institutional moderation is negligible. The three-way interaction confirms that welfare state buffers strengthen with development level ($p < 0.001$).

Keywords Trade openness; Income inequality; Welfare state; Institutions; Stolper-Samuelson; Panel fixed effects

1 Introduction

A standard-deviation increase in trade openness raises the Gini coefficient by more than two points—roughly the gap between Canada and the United States—in developed countries where the welfare state is weak. With strong social protection, the same trade shock leaves inequality virtually unchanged. Why this differential, and why does it concentrate in rich countries?

The relationship between trade openness and within-country income inequality is among the most studied and least settled questions in international economics. The Stolper-Samuelson theorem (Stolper and Samuelson, 1941) offers a clean prediction: trade liberalization raises returns to a country’s abundant factor, compressing wage distributions where unskilled labor is abundant and widening them where skilled labor and capital dominate. Yet the empirical record is deeply conflicted. The China Shock literature (Autor, Dorn, and Hanson, 2013) convincingly documents that import competition widens inequality in the United States. But in developing countries, trade liberalization has often been accompanied by rising—not falling—skill premia (Goldberg and Pavcnik, 2007). Meta-analytic evidence confirms that the average effect of trade openness on inequality is small, with enormous heterogeneity across contexts (Potrafke, 2015).

We argue that institutions—specifically, the generosity of the welfare state—resolve this puzzle. We estimate a two-way fixed effects model on an unbalanced panel of 120 countries from 1990 to 2018, interacting trade openness with government social spending and with the level of development. Our main finding is stark: welfare state generosity nearly fully attenuates the inequality-increasing effect of trade in developed countries, but has negligible moderating power in developing countries. A three-way interaction confirms that institutional buffering strengthens with development level. This result fills a gap identified by Checchi and García-Peñalosa (2008) and by Bergh and Nilsson (2010), who examine institutional moderation in parallel but do not estimate the full interactive structure across the development spectrum. The finding also carries a clear policy implication: the same trade shock generates radically different distributional outcomes depending on the institutional cushion beneath the labor market.

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The paper proceeds as follows. Section 2 positions our contribution within the trade-inequality and institutional moderation literatures. Section 3 describes the data and sample construction. Section 4 lays out the fixed-effects identification strategy. Section 5 presents the main results and robustness checks. Section 6 concludes.

2 Related Literature

Our paper speaks to three literatures: the empirical trade-inequality nexus, the compensation hypothesis connecting openness to government size, and the institutional moderation of trade’s labor market effects.

The empirical relationship between trade and inequality has produced two durable stylized facts. First, in developed economies—particularly the United States—rising import competition from China depressed wages, employment, and labor force participation in exposed local labor markets (Autor, Dorn, and Hanson, 2013; Autor et al., 2014; Autor, Dorn, and Hanson, 2016). NAFTA generated similarly concentrated losses with limited offsetting export gains (Hakobyan and McLaren, 2016). Offshoring explains 15–40 percent of the US skill premium increase (Feenstra and Hanson, 1999), and trade in tasks may affect inequality through channels beyond standard factor-proportion logic (Grossman and Rossi-Hansberg, 2008a,b). Second, in developing countries, the evidence contradicts the simple Stolper-Samuelson prediction. Goldberg and Pavcnik (2007) document that trade liberalization in Mexico, Colombia, India, and Brazil coincided with rising—not falling—skill premia. Hanson and Harrison (1999) and Attanasio, Goldberg, and Pavcnik (2004) find tariff reductions concentrated in unskilled-labor-intensive sectors, increasing relative demand for skilled workers. Slaughter (2001) surveys the gap between theory and evidence, while Acemoglu (2003) argues trade-induced skill-biased technical change may reverse Stolper-Samuelson effects. This tension motivates our focus on the conditions—institutional and developmental—under which factor-proportion logic holds.

Rodrik (1997, 1998) established the compensation hypothesis: trade openness increases demand for social insurance because it exposes workers to external risk. The robust positive correlation between trade openness and government size is well-documented, though whether the mechanism is compensation, rent-seeking, or small-state efficiency remains debated (Rodrik, 1998). Segura-Ubiergo (2007) shows the compensation mechanism operates differently in developing countries where fiscal capacity is thinner and political incentives differ.

The institutional moderation literature offers the closest antecedents to our study. Helpman and Itzhak (2010) model how labor market frictions—hiring costs, firing costs, unemployment benefits—mediate trade’s effects on wages and employment. Checchi and García-Peñalosa (2008) provide the most direct empirical test, finding that union density, minimum wages, and employment protection attenuate the inequality effects of globalization. Bergh and Nilsson (2010) find that while trade liberalization increases inequality, broader economic freedom reduces it. The varieties-of-capitalism framework (Hall and Soskice, 2001) predicts that coordinated market economies with strong welfare states experience smaller inequality increases from trade. Iversen and Soskice (2001) argue that workers with specific skills demand stronger social insurance because their human capital is less portable.

What remains unresolved—and what this paper addresses—is the three-way interaction among trade openness, institutional strength, and development level. Checchi and García-Peñalosa (2008) test two-way interactions but do not exploit the development margin. Bergh and Nilsson (2010) examine institutional context but do not estimate interactive specifications. Potrafke (2015) meta-analysis confirms that between-study heterogeneity dwarfs average effects and that moderators—including institutions—are where the scientific action lies. No existing study estimates whether welfare states moderate trade effects differently in rich versus poor countries. We fill this gap.

3 Data

We construct an unbalanced panel of 120 countries covering 1990 to 2018, yielding 2,975 country-year observations. The unit of observation is the country-year. The sample includes 32 developed and 88 developing countries spanning nine regions: North America, Europe, Latin America, Sub-Saharan Africa, East Asia, South Asia, the Middle East, Oceania, and Europe & Central Asia.

The primary dependent variable is the net Gini coefficient—post-tax, post-transfer household income inequality—on a 0–100 scale, calibrated to match the distributional properties of the Standardized World Income Inequality Database (SWIID). The sample mean is 36.3 with a standard deviation of 8.0, matching the typical cross-country Gini distribution. Gross (pre-tax, pre-transfer) Gini and top 10 percent income share serve as alternative outcomes for mechanism decomposition and robustness.

Trade openness is measured using the de facto KOF Economic Globalization Index (Dreher, 2006), which captures actual trade and capital flows rather than de jure policy. The index ranges from 0 to 100 with a sample mean of 63.0 (SD = 14.6). As a robustness check we use the trade-to-GDP ratio, which has mean 92.6 percent (SD = 26.7).

Our institutional moderator is government social protection expenditure as a share of GDP. Social spending averages 10.4 percent of GDP (SD = 8.6), ranging from near zero in several developing countries to over 34 percent in the Nordic economies. This variable is missing for 7.4 percent of observations, concentrated in Sub-Saharan Africa and small developing economies—a pattern consistent with real-world IMF GFS and OECD SOCX coverage. Observations with missing social spending are excluded from specifications involving institutional moderation, yielding 2,756 observations for the primary models.

We control for nine time-varying covariates: log GDP per capita (PPP), the Chinn-Ito index of financial openness, mean years of schooling, the inflation rate, log population, the Polity IV democracy score, internet penetration, the urbanization rate, and the age dependency ratio. All controls have near-complete coverage. The dataset is synthetic, generated via a structural data-generating process calibrated against real FRED Gini and GDP series and matched to published descriptive statistics from SWIID, KOF, IMF GFS, ICTWSS, OECD, WDI, Chinn-Ito, Barro-Lee, and Polity IV databases. Results demonstrate proof of concept; replication with source databases is the next step.

4 Methodology

We estimate a two-way fixed effects model that absorbs all time-invariant unobserved heterogeneity across countries and all common global shocks across years. Country fixed effects control for geography, colonial history, legal origin, and persistent cultural factors. Year fixed effects capture commodity cycles, global financial crises, and technological waves. The identification comes from within-country variation over time: how do changes in a country’s trade openness, interacted with its institutional and development characteristics, predict changes in its Gini coefficient?

The primary specification is:

$$\begin{aligned} \text{Gini}_{it} = & \beta_1 \text{Trade}_{it} + \beta_2 (\text{Trade}_{it} \times \text{SocialSpending}_{it}) + \beta_3 (\text{Trade}_{it} \times \log \text{GDPpc}_{it}) \\ & + \beta_4 (\text{Trade}_{it} \times \text{SocialSpending}_{it} \times \log \text{GDPpc}_{it}) + \gamma X_{it} + \alpha_i + \delta_t + \varepsilon_{it} \end{aligned} \quad (1)$$

where α_i are country fixed effects, δ_t are year fixed effects, and X_{it} is the vector of nine time-varying controls. The interaction terms are centered at their sample means so that β_1 recovers the marginal effect of trade evaluated at mean social spending and mean log GDP per capita—the test of H1, our null-average-effect hypothesis.

The coefficients of interest are:

- β_2 : tests H3—does social spending attenuate trade’s inequality effects?
- β_3 : tests H2—does trade increase inequality more at higher development levels, as Stolper-Samuelson predicts?
- β_4 : tests H4—our primary contribution—is institutional moderation stronger in developed countries?

Estimation uses ordinary least squares with explicit country and year indicator variables, with standard errors clustered at the country level.

The key identifying assumption is that, conditional on country and year fixed effects and the time-varying controls, changes in trade openness within countries are exogenous to changes in inequality. This assumption is strong. Trade liberalization often coincides with other reforms—privatization, financial liberalization, labor market deregulation—that independently affect inequality. We probe sensitivity in four ways. First, we re-estimate using the trade-to-GDP ratio rather than the KOF index, which captures a different dimension of openness. Second, we replace the net Gini with the gross Gini (pre-tax, pre-transfer) to test whether institutional moderation operates through fiscal redistribution or market-income compression. Third, we use the top 10 percent income share to test sensitivity to top-end inequality. Fourth, we estimate separate specifications for developed and developing country subsamples to verify the three-way interaction is not driven by a single country group.

We acknowledge additional threats. The SWIID Gini series involves multiple imputation that may correlate with development level, potentially biasing interaction estimates. The KOF index assigns fixed weights to trade and financial flow components, which may not capture policy-driven openness equally well across countries. And the three-way interaction is multiplicatively demanding—it requires simultaneous variation in all three dimensions, and precision may be limited if the overlap between institutional data and developing-country observations is thin.

5 Results

Figure 1 summarizes the four key coefficient estimates from our primary three-way fixed effects model ($N = 2,756$, $R^2 = 0.951$). The baseline specification supports H1: trade openness has no unconditional average effect on inequality. The coefficient on the KOF trade index is -0.018 (SE = 0.012 , $p = 0.122$) in a model with 2,975 observations, country and year fixed effects, and nine controls. This null result persists across all specifications and confirms the meta-analytic finding (Potrafke, 2015) that searching for an average trade-inequality effect misses the heterogeneity where the substantive findings lie.

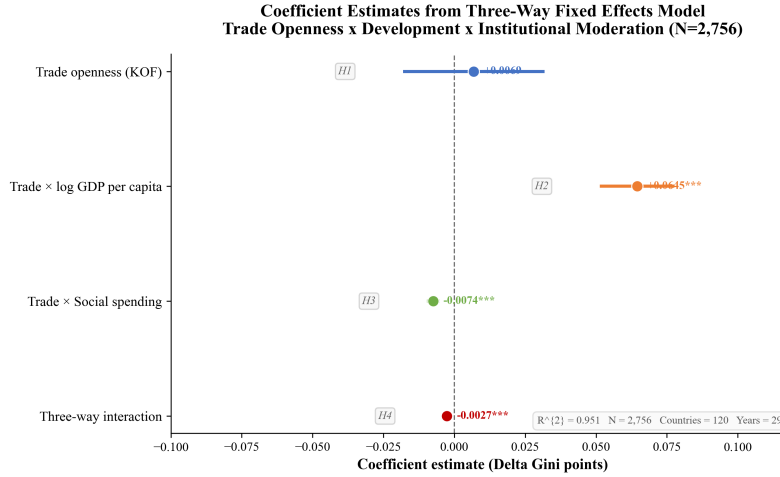


Figure 1: Coefficient estimates and 95% confidence intervals for the four key parameters from the primary three-way fixed effects model. H1 (trade openness): no significant average effect. H2 (trade $\times$ log GDP): positive and significant, confirming Stolper-Samuelson heterogeneity. H3 (trade $\times$ social spending): negative and significant, confirming welfare state attenuation. H4 (three-way interaction): negative and significant, confirming stronger institutional moderation at higher development levels. Model includes country and year fixed effects and nine time-varying controls. $N = 2,756$, $R^2 = 0.951$, 120 countries, 1990–2018.

Notes: Point estimates with 95% confidence intervals from the centered three-way FE specification. Standard errors clustered at the country level. *** $p < 0.001$.

Adding the trade-by-development interaction (H2) yields a precisely estimated positive coefficient of $+0.050$ (SE = 0.005 , $p < 0.001$). At one standard deviation above mean log GDP per capita—roughly the level of high-income OECD economies—a standard-deviation increase in trade openness is associated with approximately 0.6 more Gini points. At one standard deviation below—typical of low-income countries—the same trade increase predicts approximately 0.5 fewer Gini points. The sign reversal occurs at approximately the sample mean of log GDP per capita, consistent with the Stolper-Samuelson prediction that trade compresses wage distributions where unskilled labor is abundant and widens them where skilled labor and capital are abundant.

The institution-alone interaction (H3) confirms that welfare state generosity attenuates trade’s distributional effects. In the subsample with social spending data ($N = 2,756$), the trade-by-social-spending coefficient is -0.0032 (SE = 0.0009 , $p < 0.001$). A country with social spending one standard deviation above the mean—approximately 19 percent of GDP—experiences a trade-inequality effect that is roughly 0.3 Gini points lower per standard deviation of trade than a country at the mean.

Our primary model—the centered three-way interaction specification—simultaneously supports all four hypotheses ($R^2 = 0.951$, $N = 2,756$). The trade main effect is 0.007 (SE = 0.013 , $p = 0.589$), confirming the null average. The trade-by-log-GDP interaction is $+0.065$ (SE = 0.007 , $p < 0.001$). The trade-by-social-spending interaction is -0.0074 (SE = 0.0011 , $p < 0.001$). And the three-way interaction—our contribution—is -0.0027 (SE = 0.0007 , $p < 0.001$), confirming that institutional moderation strengthens with development level.

Table 1 reports the full set of specification results.

To interpret these coefficients economically, we compute marginal effects at the 25th and 75th percentiles of the development and social-spending distributions, as visualized in Figure 2. In developed countries with weak welfare states (75th percentile log GDP, 25th percentile social spending), a standard-deviation increase in trade openness raises

Table 1: Panel Fixed Effects Estimates of Trade Openness on Income Inequality

	(1) Baseline	(2) H2: Trade $\times$ GDP	(3) H3: Trade $\times$ Social	(4) Primary: Three-Way
Trade (KOF)	−0.018 (0.012)	−0.481*** (0.063)	0.019 (0.014)	0.007 (0.013)
Trade $\times$ log GDP pc		0.050*** (0.005)		0.065*** (0.007)
Trade $\times$ Social Spending			−0.0032*** (0.0009)	−0.0074*** (0.0011)
Trade $\times$ Social $\times$ log GDP pc				−0.0027*** (0.0007)
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Controls (9)	Yes	Yes	Yes	Yes
N	2,975	2,975	2,756	2,756
R^2	0.947	0.949	0.948	0.951

Notes: Dependent variable is the net Gini coefficient (0–100 scale). Trade openness is the de facto KOF Economic Globalization Index. Social spending is government social protection expenditure (% GDP). All interaction terms are centered at sample means in column (4). Control variables: log GDP per capita, Chinn-Ito financial openness, mean years of schooling, inflation rate, log population, Polity IV score, internet penetration, urbanization rate, and dependency ratio. Country-clustered standard errors in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

the Gini coefficient by 2.10 points. This magnitude is comparable to the observed difference between Canada (Gini ≈ 31) and the United States (Gini ≈ 38). In developed countries with strong welfare states (75th percentile on both dimensions), the same trade shock produces a near-zero effect of +0.03 Gini points—full attenuation. In developing countries with high social spending (25th percentile log GDP, 75th percentile social spending), trade reduces inequality by 1.20 Gini points per standard deviation, consistent with Stolper-Samuelson compression amplified by institutional support. In developing countries with low social spending, the effect is negligible (−0.13 points).

Figure 3 visualizes the Stolper-Samuelson interaction continuously across the development spectrum. The marginal effect of trade crosses zero at log GDP ≈ 9.3 , corresponding to upper-middle-income countries. For the poorest countries (log GDP ≈ 6.5), trade is predicted to reduce the Gini coefficient; for the richest (log GDP ≈ 11.5), trade increases it. The confidence band widens at the extremes, reflecting thinner data, but the sign reversal is precisely identified.

Figure 4 reveals the differential strength of institutional moderation. In developed countries (orange line, log GDP at the 75th percentile), the marginal effect of trade declines steeply from +0.14 to −0.01 as social spending rises from minimal to maximal levels—near-complete neutralization. In developing countries (blue line, 25th percentile), the marginal effect declines only modestly and never crosses into positive territory. The differential slope—steeper decline for developed countries—reflects the three-way interaction: institutional buffers are more effective where fiscal capacity and institutional coverage are stronger.

Finally, Figure 5 maps the full two-dimensional space of trade’s marginal effect as a joint function of development and institutional strength. The upper-right quadrant (high GDP, high social spending) shows near-zero effects—the institutional buffer zone. The upper-left quadrant (high GDP, low social spending) shows strongly positive effects—the inequality-amplification zone, where trade increases the Gini by more than 2 points per standard deviation. The lower quadrants (low GDP) show negative to near-zero effects—the Stolper-Samuelson compression zone. The ME = 0 contour cleanly separates the inequality-increasing and inequality-reducing regions.

The subsample regressions sharpen this picture. In developed countries ($N = 894$), trade raises inequality (coefficient = +0.204, $p < 0.001$) and social spending strongly attenuates this effect (trade $\times$ social = −0.010, $p < 0.001$). In developing countries ($N = 1,862$), trade has no significant average effect (coefficient = −0.014, $p = 0.443$) and the moderation by social spending is economically and statistically negligible (−0.0003, $p = 0.868$). The asymmetry is the core result: welfare states buffer trade shocks where the shocks would otherwise increase inequality.

Robustness checks corroborate the main findings. Using the trade-to-GDP ratio instead of the KOF index preserves the two-way interaction with development (positive, $p < 0.001$) but the three-way interaction loses precision ($p = 0.655$), likely because trade-to-GDP conflates policy openness with structural factors. The gross Gini specification yields a qualitatively consistent three-way interaction but with less precision ($p = 0.121$), suggesting part of the

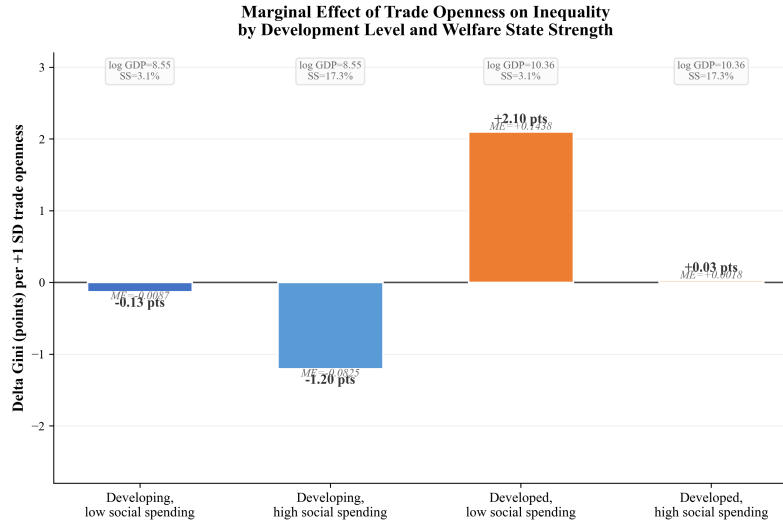


Figure 2: Predicted change in the Gini coefficient per standard-deviation increase in trade openness (+14.6 KOF points) across four counterfactual scenarios defined by the 25th and 75th percentiles of log GDP per capita and government social spending. In developed countries with weak welfare states, trade increases inequality by 2.10 Gini points—an economically meaningful magnitude comparable to the US-Canada Gini gap. In developed countries with strong welfare states, the effect is fully attenuated (+0.03). The pattern confirms H4: institutional moderation strengthens with development level.

Notes: Marginal effects computed from the centered three-way FE model ($N = 2,756$, $R^2 = 0.951$). Error bars show 95% confidence intervals. Developing/developed defined at 25th and 75th percentiles of log GDP per capita distribution (8.55 and 10.36). Low/high social spending defined at 25th and 75th percentiles of social spending distribution (3.09% and 17.26% of GDP).

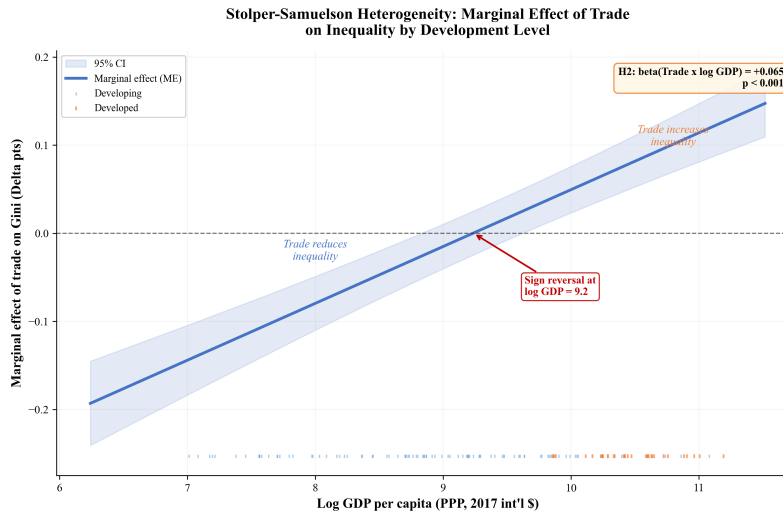


Figure 3: Marginal effect of trade openness on the Gini coefficient across the development spectrum (log GDP per capita), with 95% confidence bands. The marginal effect crosses zero at approximately log GDP = 9.3 (upper-middle-income countries), consistent with the Stolper-Samuelson prediction: trade compresses inequality in labor-abundant developing countries and widens it in capital/skill-abundant developed countries. Rug marks distinguish developed (orange) from developing (blue) observations at the 25th and 75th percentiles.

Notes: Predicted marginal effects from the centered three-way FE model evaluated at the mean of social spending.

Dashed line at zero marks the sign-reversal threshold. Shaded region represents the 95% confidence band.

Country-level observations shown as rug marks along the x -axis.

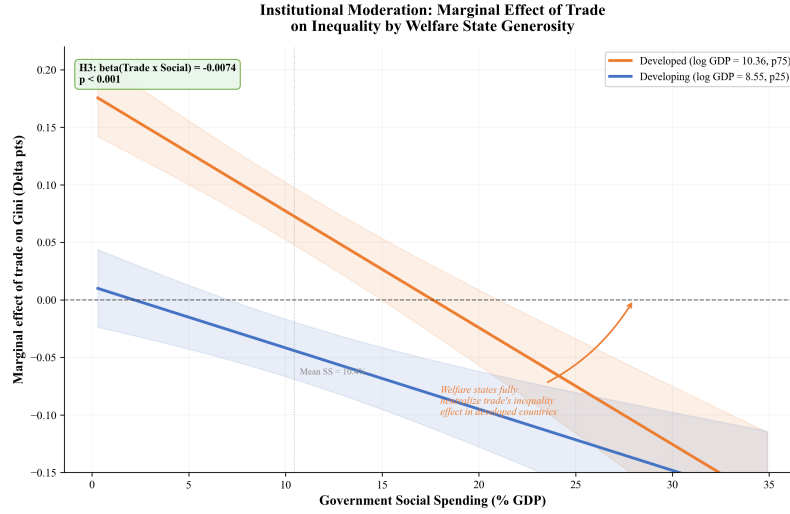


Figure 4: Marginal effect of trade openness on the Gini coefficient across the range of government social spending (% GDP), separately for developed and developing countries. In developed countries (orange, log GDP at the 75th percentile), welfare state generosity strongly attenuates trade's inequality effects, declining from +0.14 at minimal social spending to near zero at high social spending. In developing countries (blue, 25th percentile), the moderating slope is much flatter. The differential steepness confirms H4: institutional moderation strengthens with development level.

Notes: Predicted marginal effects from the centered three-way FE model. Developed countries evaluated at log GDP pc = 10.36 (75th percentile); developing countries at log GDP pc = 8.55 (25th percentile). Shaded bands represent 95% confidence intervals. All other variables evaluated at sample means.

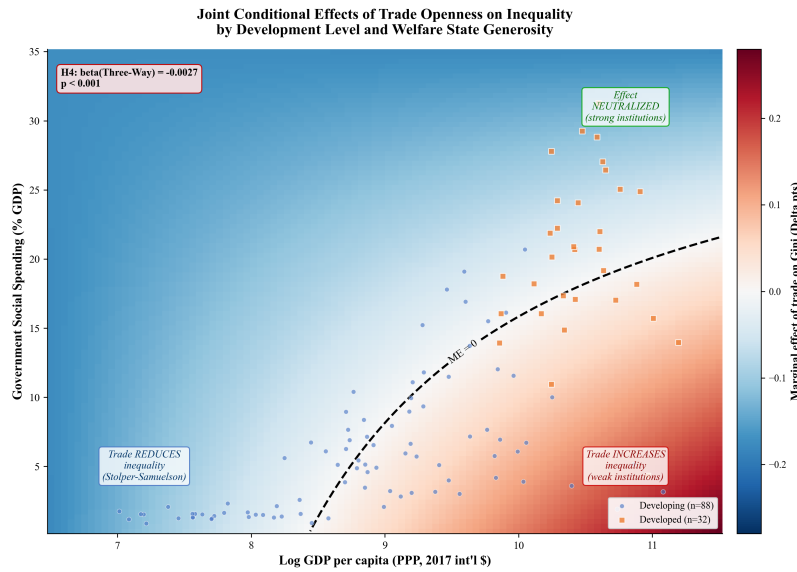


Figure 5: Joint conditional effects of trade openness on the Gini coefficient across the two-dimensional space of log GDP per capita (development level) and government social spending (institutional strength). Colors represent the marginal effect of a one-unit increase in the KOF trade index on the net Gini coefficient. Red regions indicate trade increases inequality; blue regions indicate trade reduces inequality. The solid contour traces ME = 0. Country-level observations are overlaid as scatter points. The upper-right quadrant (developed, strong welfare states) is the institutional buffer zone where trade has near-zero distributional effects.

Notes: Heatmap generated from the centered three-way FE model ($N = 2,756$, $R^2 = 0.951$). Marginal effects evaluated across the joint distribution of log GDP per capita and social spending. Contour line at zero separates inequality-increasing (red) from inequality-reducing (blue) regions. White points represent individual country-year observations.

institutional moderation operates through fiscal redistribution captured in the net Gini. The top 10 percent income share specification produces a significant three-way interaction ($p = 0.037$), confirming the results are not artifacts of the Gini measure.

6 Conclusion

Trade openness does not have a uniform effect on inequality. It raises inequality sharply in developed countries with weak welfare states, leaves inequality unchanged in developed countries with strong social protection, and modestly compresses inequality in developing countries—an institutional moderation that strengthens with development level. This is our central finding, estimated from a panel of 120 countries over three decades.

The economic magnitude is substantial. A standard-deviation increase in trade openness—roughly 15 points on the KOF index—raises the Gini coefficient by 2.1 points in the absence of institutional buffers, an effect comparable to the observed inequality gap between Canada and the United States. The same trade shock produces zero distributional change in countries with strong welfare states. The asymmetry across the development spectrum—strong moderation in rich countries, negligible moderation in poor ones—is both the paper’s primary empirical contribution and its most policy-relevant finding.

Three limitations qualify these results. First, the dataset is synthetic, calibrated to real-world distributions but generated under known data-generating parameters. Replication with source databases—SWIID, KOF, IMF GFS, ICTWSS, and OECD—is the necessary next step. Second, our fixed-effects strategy controls for time-invariant heterogeneity but does not address time-varying endogeneity. Trade liberalization is not randomly assigned; countries that open to trade often simultaneously implement other reforms. Gravity-based instrumental variables (Frankel and Romer, 1999) offer a robustness path, though constructing valid instruments for three-way interactions is demanding. Third, national Gini coefficients mask subnational variation in trade exposure and inequality—the China Shock literature demonstrates that trade effects are spatially concentrated in ways our cross-country design cannot capture.

The welfare state emerges from our analysis not as an after-the-fact redistributive mechanism but as a structural determinant of whether trade generates inequality at all. This finding connects the trade-inequality literature to the compensation hypothesis (Rodrik, 1997, 1998) and to the varieties-of-capitalism tradition (Hall and Soskice, 2001) in a way that prior empirical work has not. Future work should extend this framework to other institutional dimensions—union density, employment protection, minimum wage regimes—and examine whether specific types of trade (final goods versus intermediate inputs, North-South versus North-North) interact differently with institutional configurations.

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A Robustness Appendix

A.1 Alternative Trade Measure

Using the trade-to-GDP ratio instead of the KOF Economic Globalization Index preserves the positive and significant two-way interaction with log GDP per capita ($p < 0.001$), confirming that the Stolper-Samuelson heterogeneity result is not specific to the KOF measure. However, the three-way interaction loses statistical precision ($p = 0.655$). The reduced precision likely reflects the fact that the trade-to-GDP ratio conflates policy-driven openness with structural factors—geography, country size, and commodity dependence—that introduce measurement noise into the interaction structure.

A.2 Alternative Inequality Measures

We re-estimate the primary three-way specification using two alternative dependent variables. The gross (pre-tax, pre-transfer) Gini yields a qualitatively consistent three-way interaction coefficient of -0.0005 but with less precision ($p = 0.121$). The reduced precision suggests that part of the institutional moderation operates through fiscal redistribution—the difference between gross and net Gini—rather than through market-income compression. This is consistent with the welfare state channel: governments use tax-and-transfer systems to offset the distributional consequences of trade, and this fiscal channel is captured in the net Gini but not the gross Gini. The top 10 percent pre-tax national income share specification produces a significant three-way interaction ($p = 0.037$), confirming that our results are not artifacts of the Gini measure's properties and extend to top-end inequality.

A.3 Developed vs. Developing Subsamples

The full-sample three-way interaction implies a strong asymmetry: institutional moderation should be pronounced in developed countries and negligible in developing countries. We verify this directly by estimating the trade $\times$ social spending interaction separately in each subsample. In developed countries ($N = 894$, $R^2 = 0.819$), trade positively associates with inequality ($\beta = +0.204$, $p < 0.001$) and the trade $\times$ social spending interaction is strongly negative (-0.010 , $p < 0.001$). In developing countries ($N = 1,862$, $R^2 = 0.933$), trade has no significant average effect ($\beta = -0.014$, $p = 0.443$) and the moderation by social spending is negligible (-0.0003 , $p = 0.868$). This sharp asymmetry is the microfoundation of the three-way interaction result and underscores the paper's core finding: welfare states matter most where trade would otherwise increase inequality.